Lateral analysis – Level 3 – Forced response analysis – OH and BB pumps

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1. Purpose

This document is related to the dynamic analysis requirements for rotodynamic pump, especially to lateral analysis.

The procedure to define what level of detail for the dynamic analysis is required to be in accordance with good practice of pump design is described in *G.10.00.09* and *ANSI/HI 9.6.8-2014*: if calculated RUN is higher than 50 and less equal to 160, a **Level 3 – Advanced method – Lateral rotordynamic analysis – Forced response analysis** is required**.**

This document describes how to perform a forced response (lateral) analysis including unbalance, misalignment, and vane pass excitations.

2. Scope

This procedure is applicable to OH or BB pumps regardless of bearing type.

The procedure is applicable to the plant of Nova Milanese and the plant of La Spezia.

3. Definitions

AFB: axial fluid film bearing

BAS: base excitation

BB: between bearings (API610)

D: damping ratio

E: eccentricity

e_{per}: permissible eccentricity

F, f: force or frequency

FEA: finite element analysis

FRB: floating ring bearing

G: balance quality grade

GUI: graphical user interface

HAF: harmonic force response

HAR: harmonic response analysis

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HBS: bent shaft excitation

HOA: angular offset excitation

HOP: parallel offset excitation

INIH: integer and non-integer harmonic excitation

M, m: mass

N, n: rotating speed

OH: overhung (API610)

R: risk number (ANSI HI 9.6.8-2014)

REB: rolling Element Bearing

RFB: radial fluid film bearing

RUN: R multiplied by U (ANSI HI 9.6.8-2014)

SAN: static analysis

U: unbalance load (API610 & ISO 1940-1)

U: uncertainty value (ANSI HI 9.6.8-2014)

UNB: unbalance

U_{per}: permissible unbalance load (API610 & ISO 1940-1)

Φ_U: phase angle unbalance

Ω: rotating speed

Campbell diagram: it indicates the natural frequencies within the excitation natural frequency range and displays the margin between potential torsional resonances identified the analysis versus those excitation frequencies (1x running speed, 2x running speed, vane pass, 1x and 2x electrical line frequency, etc…).

Critical Speed: shaft rotational speed at which the rotor-bearing-support system is in a state of resonance.

Station: boundary points of a discrete element.

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4. General procedure: description

In OH or BB pumps, a lateral rotodynamic analysis is performed to assess the potential for high vibration, associated degradation, and eventually equipment failure after installation. Excessive lateral vibration can lead, or contribute, to the following types of problems:

- Accelerated wear at internal wear ring clearances and loss of performance;
- Rubs at internal locations;
- Accelerated bearing wear;
- Excitation of high vibration associated with system structural modes, leading to possible fatigue of structural components.

Lateral critical speed analysis for OH or BB pumps is performed for the combined pump rotating components (shaft, impellers, sleeves, coupling half, etc.) up to the first flexible shaft coupling (including the half-coupling), or for the entire pump-driver rotor system if a rigid coupling is used. Because of the strong effects of seal damping in most centrifugal pumps, the damped analysis is the most relevant for most of pumping applications, particularly for pumps of more than two stages.

The entire pump rotor should be included in the model. This includes the rotating component dimensions, weights, and inertias, as well as support stiffness from bearings, and internal seal clearances. The shaft cross-sectional inertial and mass properties are calculated from the length, outer diameter, and inner diameter of each shaft section from the shaft drawing. Impellers are modelled as added masses and rotational/transverse inertias at referenced shaft locations. Gyroscopic effects are included by the action of the component polar and transverse mass moment of inertia. This is particularly important with OH pumps because the impeller produces a restoring moment when whirling in a deflected position. A flexible coupling may isolate the lateral vibration between the pump and the driver. As per API RP684 best practice, the coupling half-weight (including the spacer, if any) is modelled at the centre of gravity of the half-coupling. The mathematical representation of shaft elements should include not only bending stiffness, but also transverse shear stiffness.

Hydrodynamic bearing effective support stiffness is determined from lube oil pressure, lubricant characteristics, and bearing geometry/clearance, with the result that the bearings are represented as two-dimensional spring and damping coefficients in the model including the so-called “cross-coupled” stiffness that represents reaction force build up in the direction perpendicular to the shaft motion instead of in-line with it (“direct” stiffness). Stiffness versus load curves provided by the bearing manufacturer is used to input rolling element bearing stiffness. The impact on critical speeds,

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damping and cross-coupling from annular seals (grooved or non-grooved) and pressure-reducing devices used in multistage pumps are accounted for in the analysis. Changes in these devices can impact critical speed values and definitely have a positive or negative impact on damping. The structural support stiffness from bearing mounting components, referenced to ground, should be included if the stiffness of these components is less than 10 times the stiffness of the bearing itself.

The analyst calculates the natural frequencies and characteristics of each mode (mode shape, damping, etc.). Fluid flowing/leaking through wear rings (Lomakin effect) or the specifics of a bearing design affects the stiffness and damping of rotor system. If the natural frequency is not well damped (refer to API RP684 or API Annex I) and it is located near a potential excitation frequency (1X, 2X, vane pass, line frequency, etc.), then an iteration of the analysis should be performed such that suitable recommendations can be made to correct the issue. Iteration may involve a shaft diameter, impeller weight, coupling, or bearing geometry modification.

A lateral analysis should be performed at the design point with “as-new” operating clearances, and include an iteration considering “worn” operating clearances. As a point of reference, API 610 requires the worn clearance analysis to be performed at twice the “as-new” clearances. In addition, a specification may call for a lateral forced response analysis. This is performed by taking the completed lateral analysis and applying an expected worse case or specification maximum acceptable imbalance to the rotor to determine the amplitude of any vibration (taking into account any resonances and the predicted damping or a particular mode shape).

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5. Methodology, analysis and interpretation of results

Methodology

Software/Tools:

- specialty rotordynamic programs;

The tool described in this procedure is:

- Madyn 2000

Model

See *SOP-518 – Level 2 Damped natural frequency analysis – OH and BB pumps with more than two stages* for model information.

Nonrigid bearing support stiffness (e.g., flexible bearing housing) should be considered if deemed appropriate.

Analysis

This is performed by taking the completed lateral analysis and applying an expected worse case or specification maximum acceptable imbalance to the rotor to determine the amplitude of any vibration (taking into account any resonances and the predicted damping or a particular mode shape).

This is performed by taking the completed damped lateral analysis.

The unbalance should be applied in those locations where the worst-case response is expected. They depend on the mode being excited and should located in regions demonstrating significant amplitude in the mode shape. In general, unbalance loads are applied at lumped mass stations (impeller, coupling, …) closest to the maximum amplitude points in the mode shape.
For instance, Figure 1 on the left, represent the first mode of a between bearing rotor. The point to place an unbalance load is at the lumped mass closest to mid-span. On the other end, on the mode on the right, a mid-span unbalance load would have no effect since it would be located in proximity to a node. Instead, it should be excited by two unbalance loads, one at either end of the shaft and 180° out of phase with each other to simulate the worst case.

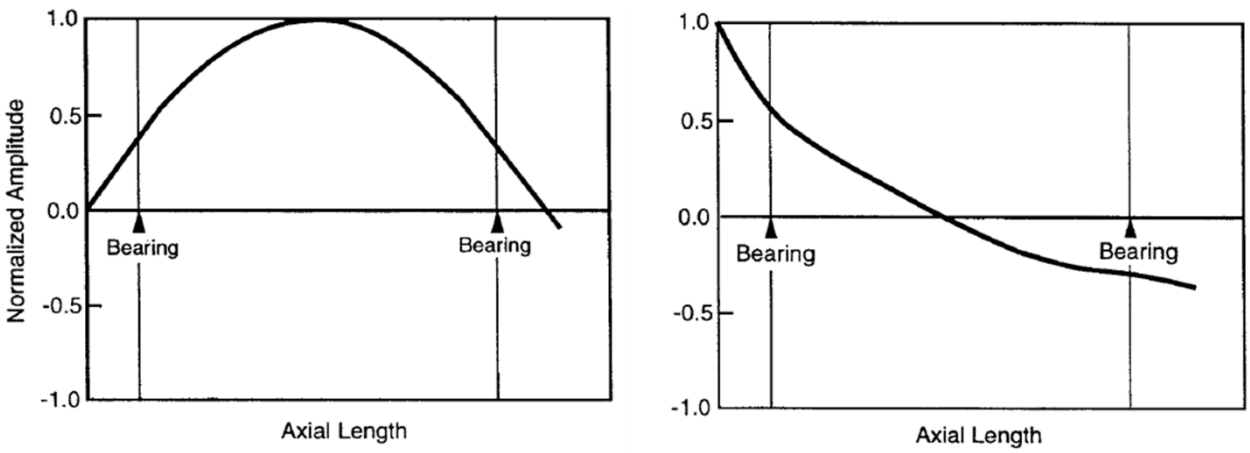

Figure 1. Mode shapes.

If the maximum amplitude position does not present lumped masses, it is unrealistic to apply a load there. Instead, it should be to components in the vicinity.

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A total unbalance of four times (4x) the allowable value (U_{per}) of the single component should be used with its balance quality grade (G).
Interpretation of results An assessment of calculated displacements versus bearing and all other close clearances between rotating and nonrotating parts shall be performed. Potential for bearing or mechanical seal damage should be assessed.

5.1. Permissible residual unbalance

The permissible residual unbalance U_{per}, according to ISO 1940-1, can be derived on the basis of a selected balance quality grade G by the following equation:

$$U_{per} = 1000 \cdot \frac{(e_{per} \cdot \Omega) \cdot m}{\Omega} = 9549.3 \cdot \frac{G \cdot m}{\mathrm{N}}$$

Where:

- U_{per} is the numerical value of the permissible residual unbalance, [g*mm]
- e_{per}*Ω (or G) is the numerical of the selected balance quality grade, [mm/s]
- m is the numerical value of the component mass, [kg]
- Ω (or N) is the numerical value of the angular velocity of the service speed, [rad/s or rpm]

API formula is approximately equal to the midpoint of the corresponding ISO balance quality grade, and it is very restrictive (Table 1).

Table 1. Residual unbalance.

G	ISO 1940	API 610
0.67	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 6350 \times \frac{m}{N}$
1.0	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 1.5 \times 6350 \times \frac{m}{N}$
2.5	$U_{per} = 9549 \times G \times \frac{m}{N}$	$U_{per} = 4 \times 6350 \times \frac{m}{N}$

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6. Acceptance criteria

As indicated in API 610 12th Edition Annex I,

- if the damping factor versus separation margin for a mode or modes is not acceptable (*SOP-518 – Level 2 Damped natural frequency analysis – OH and BB pumps with more than two stages)*
- and it is not possible to correct the issue with changes such as shaft diameter, impeller and/or coupling weight, bearing and/or wear ring geometry modification,

a rotor's damped response to unbalance shall be determined for the mode(s) in question.

A total unbalance of four times (4x) the allowable value lumped at one or more points should be investigate.

The peak-to-peak displacement of the unbalanced rotor at the point(s) of maximum displacement shall not exceed 35% of the diametral running clearance at that point.

$$displacement\ pk - pk \leq 35\%\ diametral\ clearance$$

Figure 2 summarizes the force response analysis procedure.

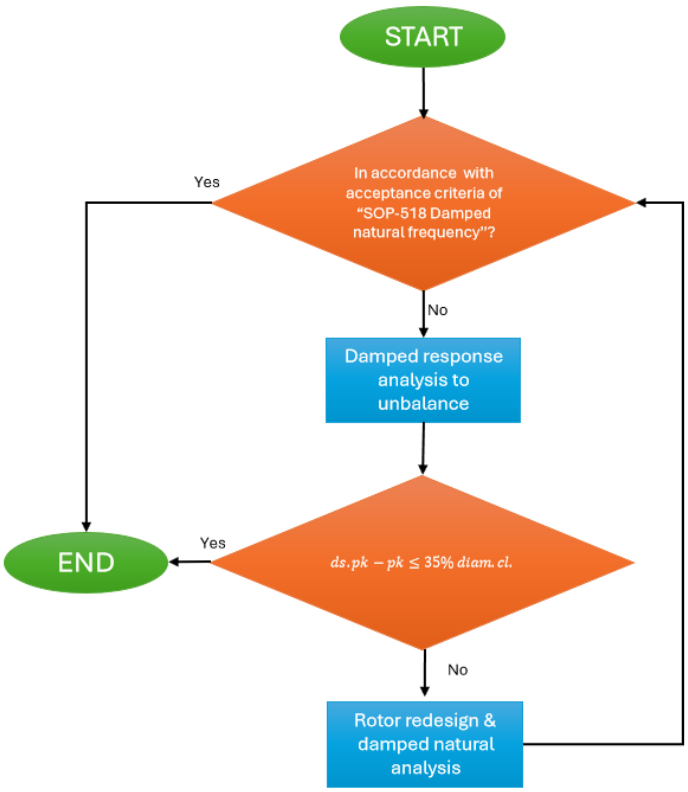

Figure 2. Forced response analysis flowchart.

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7. References

- *API STANDARD 610, 12th Edition, January 2021*
- *API PUBBLICATION 684, 1st Edition, February 1996*
- *ANSI/HI 9.6.8-2014 Rotordynamic Pumps – Guideline for Dynamic Pumping Machinery*
- *ISO 1940-1:2003 Mechanincal vibration – Balance quality requirements for rotors in a constant (rigid) state*
- *Madyn 2000 Documentation, August 2023*
- *Corbo, Mark A., and Stanley B. Malanoski. "Pump rotordynamics made simple." Proceedings of the 15th International Pump Users Symposium. Texas A&M University. Turbomachinery Laboratories, 1998.*
- *G.10.00.09 - Guidelines for dynamic analysis complexity level*
- *SOP-518 – Level 2 Damped natural frequency analysis – OH and BB pumps with more than two stages*

8. Annex A - Forced response analysis with MADYN 2000

8.1 Madyn 2000, Harmonic Response Analysis HAR

The harmonic response can be calculated for several load types and types of excitations. Excitations can be synchronous (corresponding to the shaft speed, where the load is applied), non-synchronous at one speed and non-synchronous with excitation frequencies as a function of speed.

Base excitation (BAS) is defined by an acceleration vector with 3 components in the 1,2,3 direction and a time phase angle for each direction.

Unbalance and unbalance moment loads (UNB), Figure 3.

- The magnitude can either be defined:
  - directly (radio button “Define Unbalance” active), suggested method;
  - or by the balancing quality G according to the ISO standard 1940 (radio button “Define Balance Quality” active). In this case a reference mass and reference speed are necessary (G value x yields eccentricity e via $e \cdot \Omega = x$ [mm/s] with Ω as the speed in [rad/s] $\rightarrow U = m_{Reference} \cdot e$). The shaft nominal speed is taken as the reference speed and the rotor mass or the mass of selected sectors as reference mass. Sectors can be defined in a menu, which is opened by the “set”

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button under the title “Reference mass”. The complete reference mass is shown in a field.

- Load can be applied at shaft stations and mass centres of gravity, which may have an offset to the station, where the mass is fixed.
    - Loads should be applied in regions demonstrating significant amplitude in the mode shape.
- They are defined by a magnitude U and a phase angle ϕ_U for the unbalance position.
    - For the magnitude refer to the equations in 5.1.
    - The phase angle should be defined to simulate the worst-case scenario considering the mode shape.

Harmonic forces (HAF) can be applied at shaft stations and mass centres of gravity as well. They are defined by force and moment amplitudes for all 6 co-ordinates and a time phase angle Φ for each component. The excitation function for a harmonic force is as follows:

$$F = \hat{F} \cos(\omega t + \phi)$$

with ω as the excitation frequency in [rad/s].

Note, that the excitation frequency for harmonic forces ω in equation can also be rotational speeds Ω, if specified by the check box “Synchronous Excitation” when starting the analysis.

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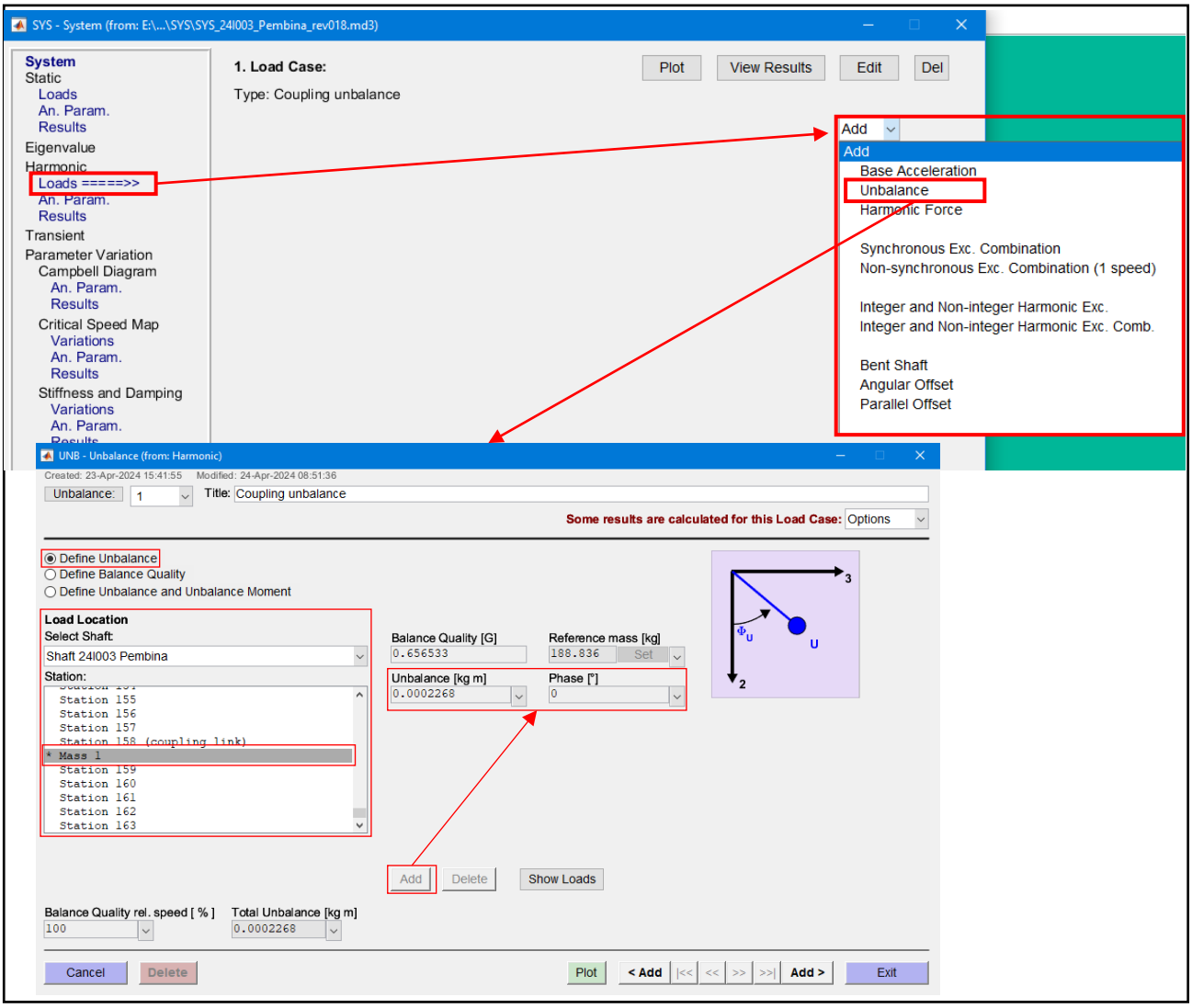

Figure 3. Harmonic Load case, Unbalance GUI

A **bent shaft** (HBS)is defined by two nodes, a maximum displacement between the two nodes caused by the bending and a phase angle for the direction of the maximum displacement. Phase angle zero means, that the lateral displacement between the nodes is in 2-direction.

The two nodes should be part of the same shaft or belong to rigidly connected shafts.

Angular and parallel offsets are meant to simulate imperfections in coupling flanges as indicated in Figure 4. Both are defined by 2 nodes, which should be the nodes at the left end of the left flange and the right end of the right flange, which have the same inner and outer diameter. The flanges must be rigidly connected.

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- An **angular offset (HOA)** is defined an angle between two nodes and a phase angle for the direction of the offset. Phase angle zero means, that the lateral displacement caused by the offset is in 2-direction.
- A **parallel offset (HOP)** is defined an offset value and a phase angle for the direction of the offset. Phase angle zero means, that the lateral displacement caused by the offset is in 2-direction.

In case the forces are not applied at two equal rigidly connected flanges as described above the applied forces might not yield the desired deformation.

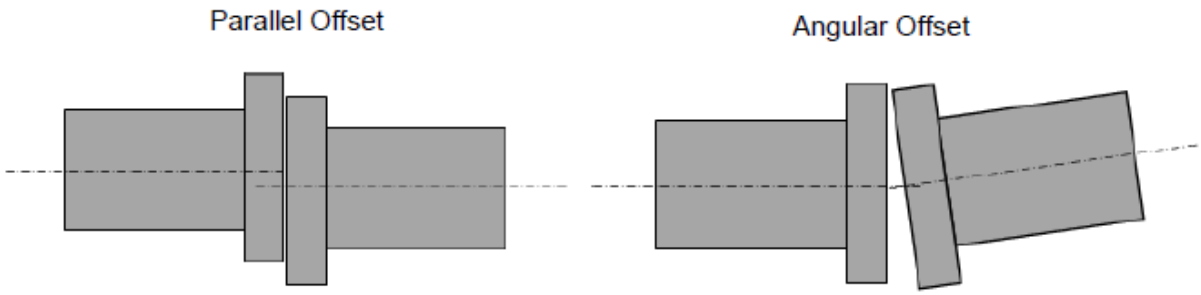

Figure 4. Offset of flange connections.

Integer and non-integer harmonic (INIH) excitations, apart from the usual inputs of forces and moments at stations, allow defining a frequency, speeds and factors i_f and i_n by which the speed *n* and the frequency *f* are multiplied and combined to define an excitation frequency according to the following formula:

$$f_e = i_n n + i_f f$$

The defined forces and moments for INIH excitations may vary with the speed. They can be multiplied with a speed function f(*n*). The speed function for each of the defined speeds in the load GUI can be input in a separate window, which is invoked by the button “add Function”.

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In the harmonic analysis (Figure 5) the following parameters and settings can be defined:

- Enabling bearings requires the selection of the source for the bearing load (SAN or direct load input). The method suggested is to use the results of a static analysis (SAN).
  NOTE. If RFB/FRB/AFB/REB loads are defined from a normal static analysis result, which is speed dependent, it is necessary that the whole speed range is covered by the static analysis.
- Different type of loads can be selected for the following type of excitation frequencies:
  - Non-synchronous excitation at one rotor speed: HAF, BAS
  - Synchronous excitation (equal to rotor speed): UNB, HAF, BAS, HBS, HOA, HOP
  - Integer and non-integer harmonics, non-synchronous exciting frequencies as a function of speed: Speed dependent forces and moments

  The type of excitation is selected by check boxes in the analysis GUI, with the non-synchronous excitation at one speed as default (all check boxes unchecked).
  In case of “synchronous” excitation the frequency is defined relative to the rotor nominal rotor speed n_0.
  In case of “non-synchronous” it is defined in dimensional form. In this case there are edit fields for the start value, the end value and the step.
  For “integer and non-integer harmonic” excitation a relative speed range and a frequency step have to be input. related. Additionally a cut off frequency can be defined, which has the effect of not further refining the step, if the frequency is below this limit.
  The speed step and frequency step can be defined in two ways:
  a. Definition as a geometric row by defining Δn/n and Δf/f, respectively, which is the default option. This has the advantage, that the necessary step can be easily derived from the damping ratio D of resonances. In case of Δn/n=2D the value of Δn corresponds to resonance peak width at the half power point. An adequate Δn therefore is Δn=D with D as the minimum damping of all modes within the speed range. The same applies for Δf.
  b. Definition as a linear row by defining Δn/n0 and Δf, respectively. This option is activated by a check box. In case of a resonance at very low speeds an adequate Δn/n0 and Δf, respectively, is very small. This definition therefore will normally yield a unnecessary large number of speed steps.

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For an unbalance response analysis, the "Synchronous excitation" check box should be activated and the speed range should go from $0.25 \cdot \frac{n_{min}}{n_0}$ to $1.25 \cdot \frac{n_{max}}{n_0}$. A value of Δn/n equal to 0.005 should be sufficient for most cases.

- An additional structural damping can be defined. The following damping matrix then is added to the original damping matrix.
  A value of 1% is a reasonable number for these systems.

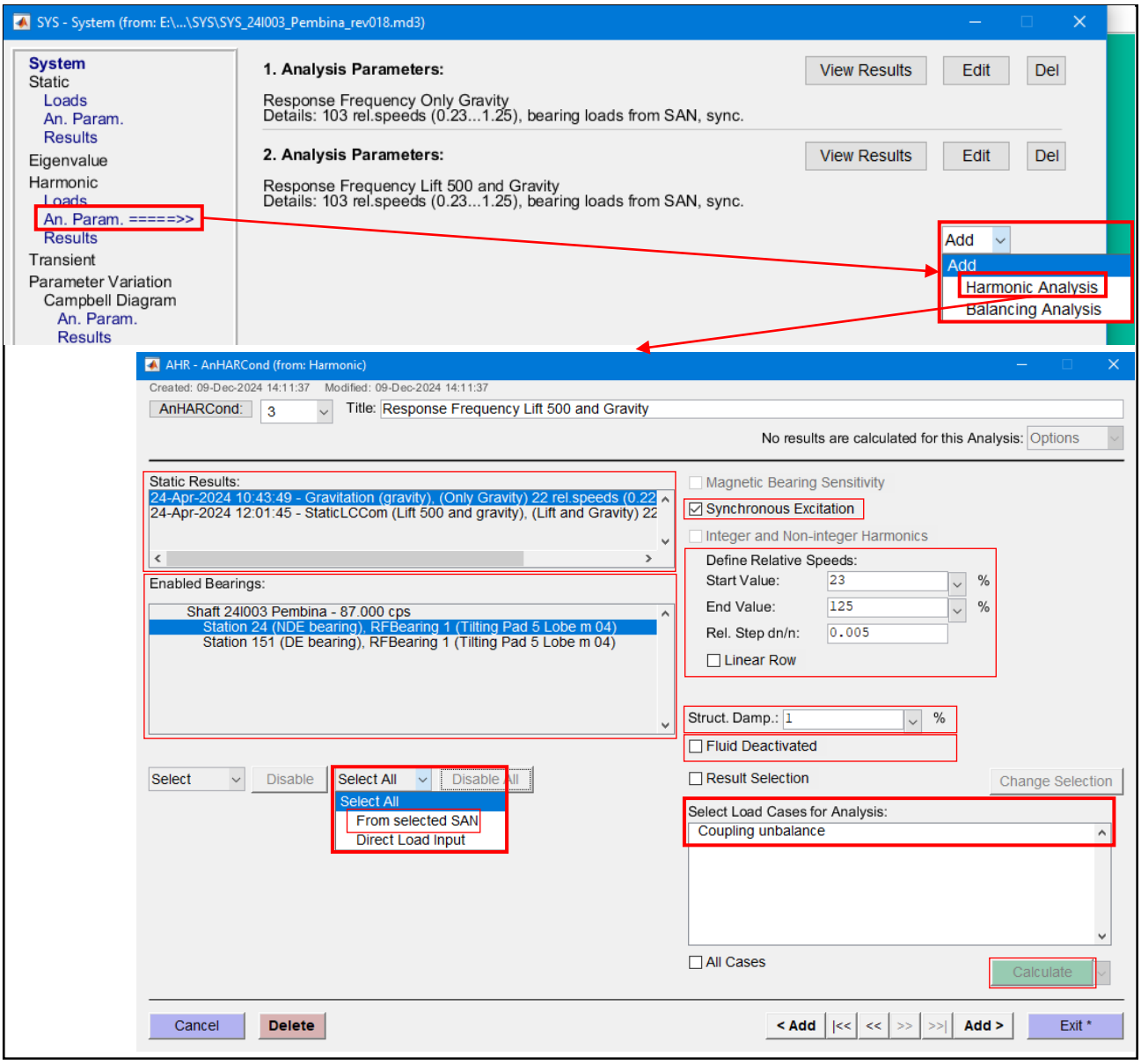

Figure 5. Harmonic Analysis, Unbalance response GUI.

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After performing the harmonic analysis, access the results as shown in Figure 6. Select the proper load case and plot the resonance plot, the shape plot or bearing coefficient plot of selected result. In all shape plots the contour of the shaft is drawn in the background.

In the harmonic analysis results are plotted as:

- Resonance Plot that shows results of different stations, such as displacement, stress, forces, …, as a function of rotating speed;
- Shape Plot that shows results, such as displacement, forces, …, along the shaft at a fixed rotating speed;
- Coefficient Plot that shows bearing coefficients (stiffness and damping) as a function of rotating speed.

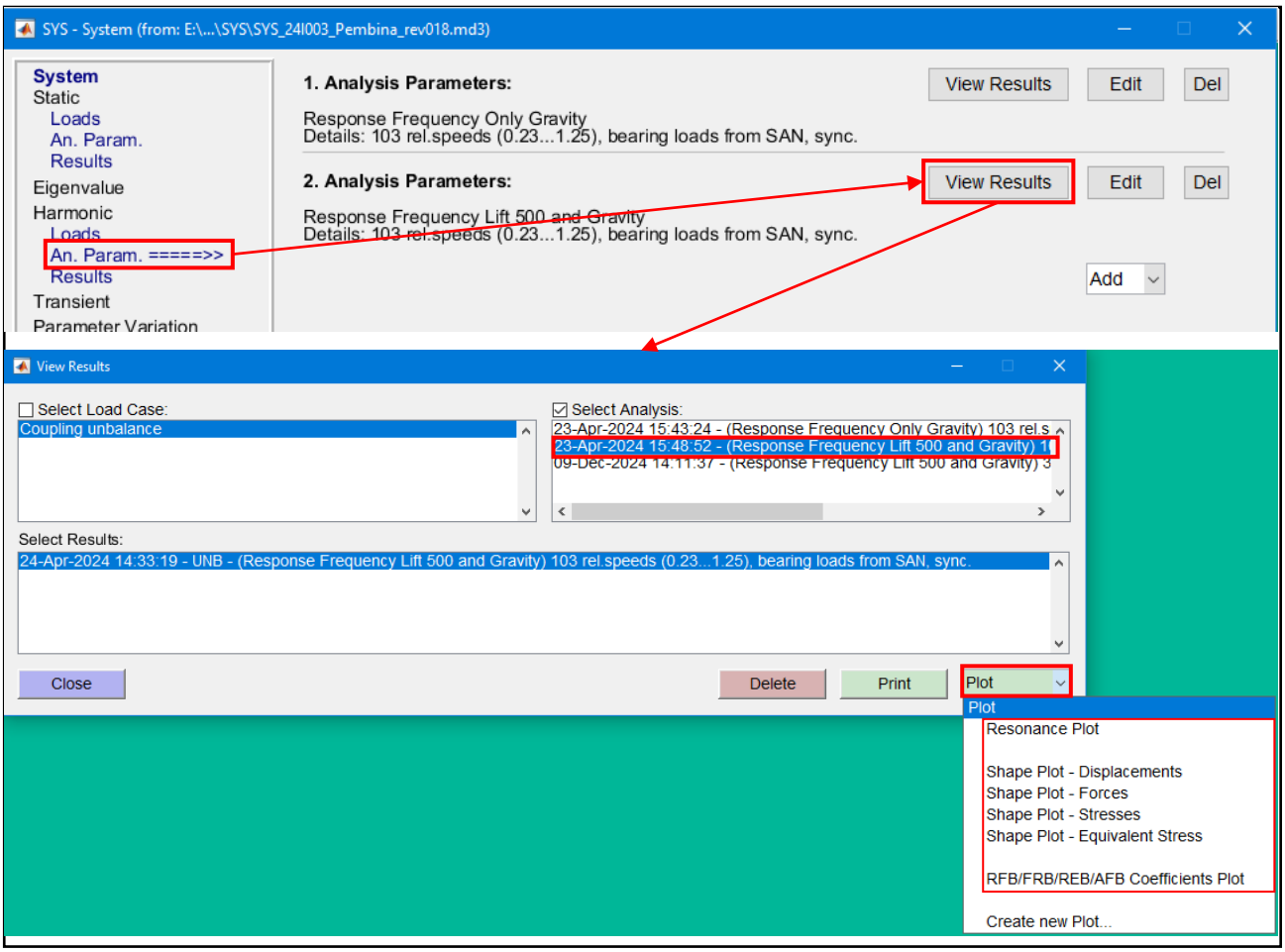

Figure 6.Harmonic Results GUI.

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